

A Data-Driven "One Health" Approach to Zero Rabies and Animal Bite in Quezon City

Bonnin, J.J., Cababasay, Y., Ching, J.M., Rivera, G.F., Rosas, R.G., Sembrano, K.

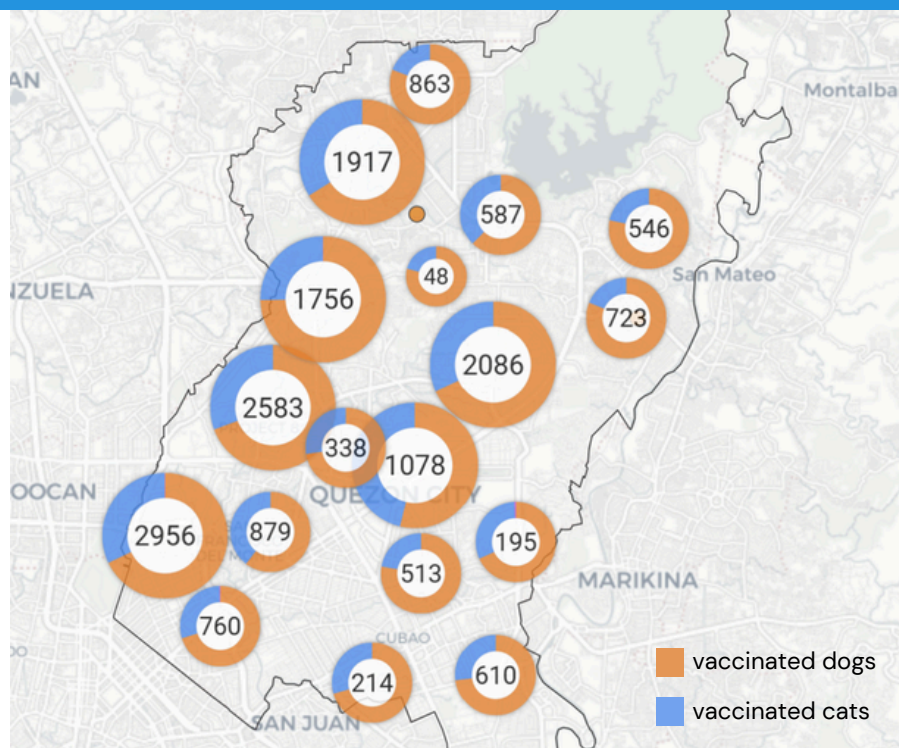
KEY MESSAGES

- Resurgent cases of rabies and animal bites are evident nationwide, with Quezon City accounting for the highest burden in NCR despite its strong budget and LGU capacity.
- Existing control strategies consist mainly of vaccination of pets against rabies; however, strategies targeted towards strays and canine population control need reassessment.
- Surveillance, animal population control, and community awareness remain fragmented, which highlights the urgent need for an integrated One Health approach that unifies data and strengthens multi-sector coordination.
- Mapping of local real-time data dynamics on human bite incidence, confirmed rabies cases, vaccination coverage, sterilization/CNVR data, animal bite treatment center (ABTC) locations, and stray density is a cornerstone for successful tracking and management.

EXECUTIVE SUMMARY

Despite the mandate of the **Anti-Rabies Act of 2007 (RA 9482)** and local control efforts, Quezon City is facing a **critical resurgence in animal bites and rabies threats**. In November 2024, the Department of Health (DOH) confirmed 15 rabies deaths in Metro Manila, identifying Quezon City as a key affected area.^[3] Local data reinforces this alarming trend, with partner hospitals reporting over **2,627 animal bite cases in 2024** alone,^[10] contributing to the city's staggering total of **138,102 cases—the highest absolute burden** in the National Capital Region.^[22]

This persistence highlights a **systemic failure**: current interventions are **reactive** and **fragmented**, lacking the cross-sectoral coordination needed to manage the rising stray population. To address this, this policy brief recommends the institutionalization of the **"Quezon City One Health Data Integration Ordinance."** This measure establishes a mandatory, real-time data-sharing protocol between the City Health Department and City Veterinary Office, creating a unified **Geographic Information System (GIS)** to transform rabies control from a siloed response into a **proactive, data-driven operation**.

AT A GLANCE**Number of Vaccinations for Dogs and Cats in Quezon City**

INTRODUCTION

Animal bites are a major public health concern in the Philippines and globally, serving as the primary pathway for the transmission of rabies. Rabies is a vaccine-preventable, zoonotic viral disease caused by the virus *Rabies lyssavirus* (RABV).^[27] It is transmitted through the saliva of infected animals, most commonly via a bite.^[14] Once clinical symptoms appear, rabies is almost always fatal.^[14]

Globally, an estimated number of 70,000 annual deaths occur due to rabies,^[7] with 99% being transmitted by dogs.^{[4][7]} Animal bite incidents, which serve as critical indicators of rabies exposure risk, has also shown a concerning rise.

70,000

annual deaths occur due to rabies globally.^[8]

NATIONAL SITUATION

In the Philippines, animal bites have consistently ranked among the top causes of morbidity: 8th in 2019, 4th in 2020, and 3rd from 2021 to 2024. In 2024 alone, cases surged by 64.31%, signaling worsening exposure risks.^[10]

Rabies morbidity likewise showed one of the highest percent increases between 2023 and 2024.^[10] Despite the Anti-Rabies Act of 2007 (RA 9482) and existing local strategies, human rabies deaths continue to rise, reaching 260 cases from January to September 20, 2025. Notably, 95% of these infections were linked to unvaccinated or unknown-status animals.^[20]

ANIMAL BITES IN THE PH

(Morbidity rank and surge)



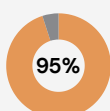
2019: Ranked 8th

2020: Ranked 4th

2021-2024: Ranked 3rd

RABIES CASES IN THE PH

260 CASES (Jan–Sept 2025)



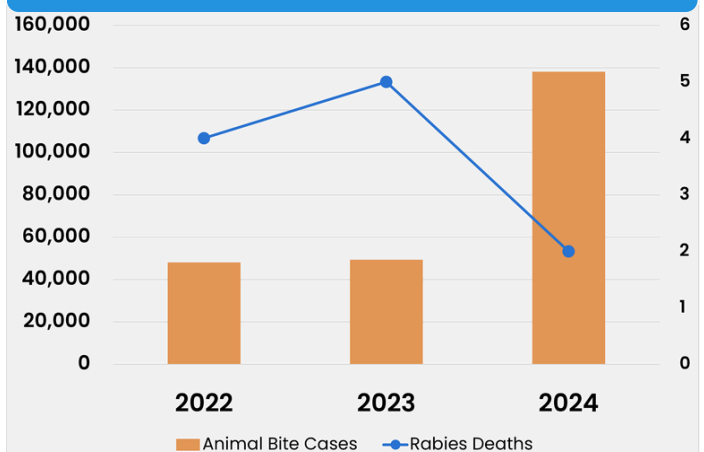
Unvaccinated/Unknown-Status animals

LOCAL SITUATION

Emerging rabies and animal bite challenges are particularly evident in one of the Philippines' biggest and most acclaimed cities: **Quezon City**. While local initiatives exist (e.g., QC's weekly spay and neuter programs aimed at population control in barangays), the rate of abandonment and reproduction often outpaces these interventions.^[12] Consequently, the number of strays remains high. These animals are often unvaccinated and prone to roaming, contributing to both rising bite cases and persistent rabies transmission risks in communities.^[12]

The city consistently records the highest burden in NCR, reporting 48,061 animal bite cases in 2022 with four rabies deaths, and 49,388 cases in 2023 with five deaths, placing it first in the NCR cities most affected.^[10] In 2024, the situation further escalated: NCR reported 461,862 cases, and Quezon City accounted for a substantial portion of these cases, with 138,102 recorded animal bites.^[22] This constitutes roughly 30% of all cases in NCR.

ANIMAL BITES AND RABIES DEATHS IN QUEZON CITY, 2022-2024



Source: 2022 & 2023: [10]; 2024: [22]

Local hospitals, such as Capitol Medical Center, reported treating over 2,627 new bite cases in 2024 alone, a number that continued to rise with 427 new cases in just the first two months of 2025.^[9] Along with this, even with the free anti-rabies vaccinations that are also provided to pets, the control of rabies in Quezon City is still subpar. Specifically, the city logged the highest number of rabies cases in a referral hospital in Manila.^[15]

Despite low rabies mortality, the data point to a system under strain. Deaths are being averted mostly through reactive post-exposure treatment, costing an estimated ₱100 million annually,^[23] rather than through sustained prevention. This reflects operational fragmentation, with bite management and animal control handled by separate departments with limited coordination.

These trends reveal a disconnect between the goal of RA 9482 and the reality that Quezon City faces, highlighting the urgent need for improved policies that aim to strengthen rabies control, surveillance, and prevention efforts in the city to reduce the incidence of animal bites and rabies cases.

THE GAPS

The escalating figures highlight **several critical policy and implementation gaps** that continue to **drive the rise in animal bites and rabies cases** in the country.

1. The predominance of bites from unvaccinated animals indicates weak enforcement of mandatory pet vaccination.
2. The rapid growth of stray animal populations indicates insufficient and inconsistently resourced spray-neuter and population management programs, which fail to keep pace with the rate of abandonment and reproduction.

Current policies only emphasize penalizing abandonment under the Republic Act No. 10631.

3. Abandonment of pets due to financial constraints or other reasons reflects a gap in responsible pet ownership education.
4. The continued occurrence of preventable rabies infections underscores gaps in public awareness and education regarding proper wound care, immediate consultation, and post-exposure prophylaxis.

Animal bites and rabies will continue to threaten human life and strain the economic capacity of Quezon City's LGU if the **canine population remains unabated**. Suggested policy recommendations that can avert such a scenario include:

POLICY RECOMMENDATION

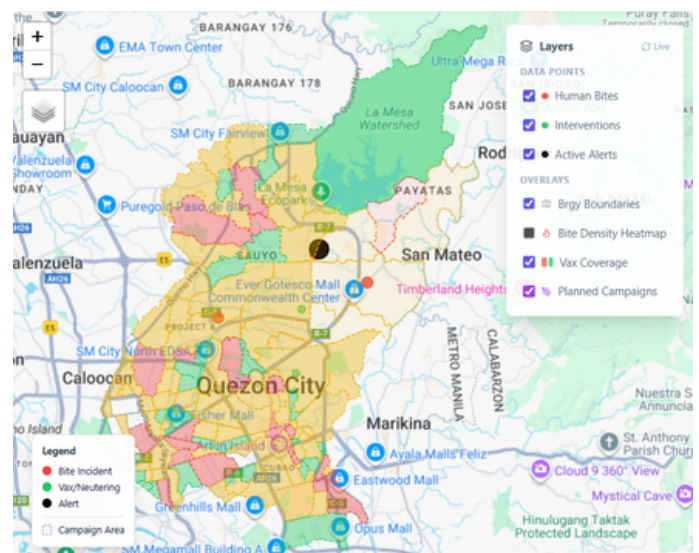
1. Establish an integrated One Health Data System and a multi-sector coordinating committee for rabies prevention in Quezon City

- **Data-Driven "One Health Approach".** To strengthen rabies prevention, Quezon City should integrate data from the City Health Department, City Veterinary Office, and private veterinary clinics into a unified Automated Geographic Information System (GIS). Using the Rabies Case Surveillance (RCS) model, the system will provide real-time tracking of human bite incidence, confirmed rabies cases, vaccination coverage, sterilization/CNVR data, animal bite treatment center (ABTC) locations, and stray density. This will enable proactive hotspot detection and more efficient planning of vaccination, sterilization, and information campaigns.

- **Improve planning and resource utilization.** The integrated database will also enhance planning and resource allocation, enabling more targeted, non-duplicative programs across the city. A unified system prevents overlapping activities, improves the scheduling of vaccination and sterilization campaigns, and enables the timely deployment of resources in high-risk areas. Decentralizing data entry to ABTCs, health units, and veterinary clinics will reduce delays caused by paper-based reporting and strengthen localized monitoring and rapid response.
- **Strengthens multi-sector response.** To institutionalize coordination, the city should establish the Quezon City Rabies Control Coordinating Committee (QCRCCC)—a multi-sector body composed of the LGU, City Health Department, City Veterinary Office, DepEd, PNP, and DA. This committee will align public health, veterinary, education, and enforcement efforts under a One Health framework.^[25]

Evidence

1. **National systems** (PhilAHIS, NaRIS) face issues of incomplete data and delayed reporting, underscoring the need for enhanced local surveillance.^[6]
2. **Digital reporting** improves real-time accuracy compared to paper-based systems.^[1]
3. **Global Alliance for Rabies Control's (GARC) RCS tool** demonstrates global success in automated mapping and outbreak tracking (e.g., Algerian Organization Against Rabies in Algeria).^[13]
4. **Multi-sector rabies councils** in Bohol, Zamboanga City, Tabaco City, and Cebu Province show improved coordination, surveillance, and reductions in human and canine rabies cases.^{[5][8][19][26]}



The QC PROACT Command Center is a functional web-based prototype designed to overlay human bite incidence with animal vaccination coverage on a real-time map, transforming raw data into actionable "Heat Maps," to aid the deployment of targeted interventions where most needed.

2. Adopt a City-wide Catch-Neuter-Vaccinate-Return (CNVR) to complement existing spay-neuter and impounding services

- **Lowers bite risk.** CNVR stabilizes free-roaming dog and cat populations and reduces nuisance behaviors associated with roaming and territorial competition. It is also associated with fewer dog bites in areas with high CNVR coverage.
- **Sustains vaccination coverage.** Sterilization reduces turnover, maintaining higher long-term rabies vaccination coverage and lowering human rabies risk.
- **More humane and cost-effective long term.** Compared to repeated impounding and culling, which are reactive measures, CNVR reduces shelter overload, repeated recapture costs, and PEP expenditure when implemented at scale. Moreover, it also improves public perception of and care provided to dogs, while reducing risks to public health, wildlife, and livestock.

Evidence

1. **Cebu City, Philippines:** Full CNVR implementation in Cebu has reported marked reductions in euthanasia and a 32-percent drop in rabies cases following rollout from January–June 2025.^[19]
2. **Greater Bangkok, Thailand:** More rounds of CNVR in Greater Bangkok have led to greater localized declines in free-roaming dogs (FRD) density, reductions in rabies cases, lower odds of reporting a bite, and improvement of dog-human relationships.^[16] The most recent unpublished analysis showed a 60% decrease in FRD over eight years of performing CNVR.^[16]
3. **India (Tamil Nadu):** A cost-effectiveness analysis found that combining female sterilization with vaccination produced better long-term reductions in stray dog populations at lower accumulative cost compared to other approaches.^[21]

3. Strengthening the Authority, Capacity, and Sustainability of Barangay Rabies Control Committees (BRCCs)

- **Localizes enforcement authority.** Authorizing BRCC from a monitoring body to an enforcement unit. Providing them with the legal authority and standard citation tickets to issue fines for violations of the Anti-Rabies Act creates a localized revenue stream that can be reinvested into committee operations.
- **Ensures operational sustainability.** Linking BRCC performance to incentives through an annual award system prevents commitment dormancy and motivates long-term enforcement.
- **Strengthens BRCC capacity and support.** Provide regular training on dog handling, data reporting, rabies education, and enforcement of regulations, and establish a direct hotline with the City Vet and City Health Department to ensure guidance, feedback, and resources when needed.

1. **Bohol City, Philippines:** The Bohol Rabies Prevention and Elimination Project proved that empowering barangays to collect mandatory dog registration fees was the primary driver of program sustainability. By allowing barangays to retain a 50% share of the fees for their own operations, the project created a self-sustaining revenue stream that funded local enforcement.^[18]
2. **Metro Manila (Caloocan City):** The "Best Barangay Rabies Program Implementer" awards successfully incentivized 17 barangays to sustain their own vaccination and education drives, proving that recognition and competitive incentives drive local compliance and program execution.^[24]
3. **Albay Province, Philippines:** Implementation of an Integrated Bite Case Management (IBCM), which specifically trained local animal health workers to investigate bites and linked them directly to veterinary offices, resulted in a fourfold increase in the detection of rabid animals.^[11]

CONCLUSION

The escalating incidence of animal bites in Quezon City, compounded by climate-related factors and environmental triggers of animal aggression,^{[2][12]} presents an **urgent public health challenge that current reactive measures can no longer contain**. Without a structural shift in how the city manages surveillance, the gap between the goals of RA 9482 and the reality on the ground will continue to widen, threatening the safety of residents in both high-density communities and institutional zones. Adopting the **One Health Data Integration Ordinance is not merely an administrative upgrade but a critical operational necessity**. By **synchronizing health and veterinary data** into a single decision-making framework, Quezon City can **move beyond "treating" the problem to proactively "preempting" it**, ensuring that interventions are deployed exactly where they are needed most to reduce animal bites and achieve a rabies-free city.

"It is time to align our governance with our data. Institutionalizing QC PROACT is the necessary step to stop treating the symptom and finally solve the problem."



GIS HEAT MAP



REFERENCES